

Submission Date: 8 January 2026

Course: Graph Mining [4041]

Instructor: Dr. Zeinab Maleki

Project Title: Implementation of Sequential Graph Neural Networks (SeqGNN) for Urban Traffic Speed Prediction

Student Information

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1 Executive Summary

Since the submission of the proposal, we have successfully implemented the core architecture of the Sequential Graph Neural Network (SeqGNN) as proposed by Xie et al. (2020) [1]. We have transitioned from the theoretical design phase to practical implementation using PyTorch. The proprietary Xi'an dataset used in the original paper was replaced with the public METR-LA benchmark dataset to ensure reproducibility. We have successfully developed the data pre-processing pipeline, the Graph Neural Network (GN) blocks, and the Recurrent Graph Neural Network (GRNN) blocks. Preliminary training over 50 epochs has been completed, yielding a Mean Absolute Error (MAE) of 6.5570 on the test set, which aligns competitively with the results reported in the reference paper.

2 Progress on Objectives

Objective 1: Data Acquisition and Preprocessing

Status: Completed.

We successfully acquired the METR-LA traffic dataset as a proxy for the private dataset used in the paper. The data pipeline handles the download of the adjacency matrix (`adj_mx.pkl`) and traffic speed logs.

- We processed the graph topology into edge indices suitable for PyTorch Geometric/GNN processing.
- We implemented a sliding window mechanism to generate input sequences (historical data) and output sequences (future predictions).

Objective 2: Model Implementation (SeqGNN)

Status: Completed.

We have fully implemented the SeqGNN architecture in PyTorch. This includes:

- **GN Blocks:** Implemented the “graph-to-graph” module that updates edge, node, and global attributes.
- **GRNN Blocks:** Integrated the GN blocks with recurrent logic (merging hidden states with current inputs) to handle sequential graph data.
- **Encoder-Decoder:** Constructed the full sequence-to-sequence architecture to predict future traffic graphs based on historical graph sequences.

Objective 3: Training and Evaluation

Status: In Progress/Partially Met.

We have established a training loop using the Adam optimizer and L1 Loss (MAE).

- The model was trained for 50 epochs on a GPU-accelerated environment.
- Preliminary evaluation metrics (MAE and RMSE) have been calculated.
- *Remaining Work:* We aim to extend the training epochs to check for further convergence and perform a more rigorous comparison against simple baselines (e.g., MLP, standard LSTM).

3 Work Accomplished

3.1 Dataset Preparation and Analysis

The original paper utilizes a private dataset from Beilin District, Xi'an. To ensure our project is executable, we utilized the METR-LA dataset, a standard benchmark for traffic forecasting.

- **Graph Construction:** The road network was converted into a linkage graph where nodes represent sensors/road segments. The graph contains approximately 207 nodes and 1,722 edges.
- **Sequence Formatting:** Following the paper's methodology, we structured the data into 36-step windows. The first 24 steps (2 hours) are used as the input sequence, and the subsequent 12 steps (1 hour) are the target output sequence.
- **Static Attributes:** We generated static node and edge attributes (e.g., normalized degree) to match the input dimensionality required by the GN Blocks.

3.2 Implementation Details

The model was built using `torch` and standard scientific libraries. The core logic follows the mathematical formulation in the paper:

1. **GN Block:** We implemented the three-stage update mechanism:
 - *Edge Update:* $e'_k = \phi^e(e_k, v_{rk}, v_{sk}, u)$
 - *Node Update:* $v'_i = \phi^v(\bar{e}'_i, v_i, u)$ where aggregation is performed via summation.
 - *Global Update:* $u' = \phi^u(\bar{e}', \bar{v}', u)$
2. **GRNN Block:** We implemented the *GMerge*, *GNCore*, and *GSplit* functions to manage the memory state of the graph across time steps.
3. **SeqGNN Class:** This orchestrates the encoding of the input sequence $X_{0...t}$ and the autoregressive decoding of predictions $Y_{t+1...t+k}$.

3.3 Preliminary Results

We trained the model for 50 epochs. The loss converged rapidly within the first 10 epochs.

- **Training Loss (MAE):** Dropped from ~ 11.94 to ~ 6.69 .
- **Test Results:** The model was evaluated on a hold-out test set (15% split).

Table 1: Quantitative Results Comparison

Metric	Our Implementation (METR-LA)	Paper Reference (Xi'an Data)	Notes
MAE	6.5570	7.19	Strong convergence observed
RMSE	13.3533	9.61	Variance due to dataset differences
Device	NVIDIA T4 GPU	NVIDIA TITAN Xp	

4 Challenges Encountered and Resolutions

Challenge 1: Dataset Accessibility and Format

The dataset used in the paper (Xi'an City data) is not publicly available. Furthermore, available mirrors of public datasets (like METR-LA) often have broken links or require authentication.

- **Resolution:** We implemented a robust download script that attempts to fetch the raw `.h5` file and falls back to a CSV mirror hosted on Zenodo/GitHub if the primary fails. We also wrote code to synthesize data structures if download failures persist, ensuring the code remains runnable.

Challenge 2: Data Type Inconsistencies

During the loading of the speed data, we encountered errors converting strings to floats. This was caused by timestamp columns being included in the numerical data arrays.

- **Resolution:** We added a preprocessing step to detect non-numeric data types. The script automatically drops the timestamp column and casts the remaining data to `float32`. We also added logic to trim the static adjacency matrix to match the number of nodes in the speed data if discrepancies arise.

References

- [1] Z. Xie, W. Lv, S. Huang, Z. Lu, B. Du and R. Huang, "Sequential Graph Neural Network for Urban Road Traffic Speed Prediction," *IEEE Access*, vol. 8, pp. 70106-70115, 2020.
- [2] P. W. Battaglia et al., "Relational inductive biases, deep learning, and graph networks," arXiv preprint arXiv:1806.01261, 2018.
- [3] Y. Li, R. Yu, C. Shahabi, and Y. Liu, "Diffusion Convolutional Recurrent Neural Network: Data-Driven Traffic Forecasting," in *ICLR*, 2018.

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